At Your Fingertips: On-Demand Object Affordance Generation Framework for Cross-Spatial Object Interaction

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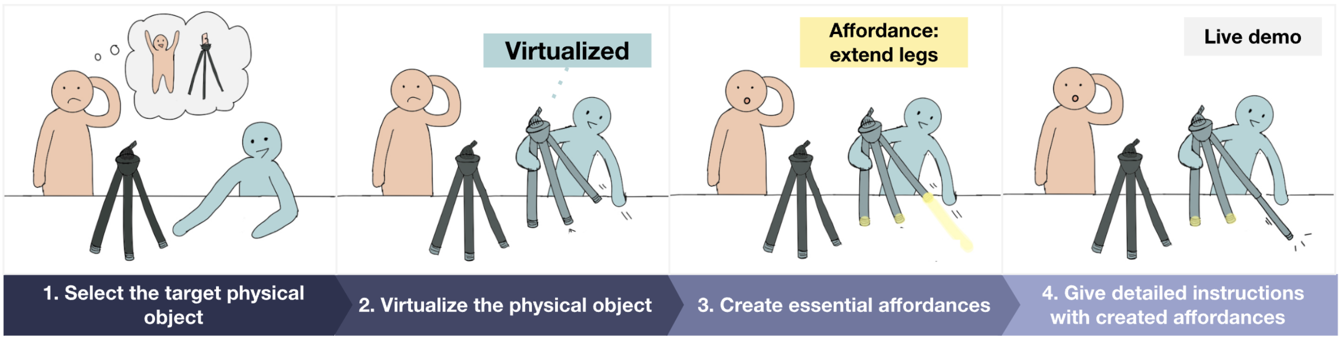

Figure 1: Example operation of CRAFT. Using the CRAFT framework, a remote expert (depicted in blue) can guide a novice user (depicted in orange) in operating complex objects, such as a camera tripod, within the novice's space. (Step 1) The expert selects the tripod as the target object. (Step 2) The system virtualizes the tripod, creating a digital representation. (Step 3) As the expert interacts with the virtual tripod leg to demonstrate how to extend it, an on-demand affordance is generated and augmented onto the virtual tripod. (Step 4) This affordance enables the novice to follow a live demonstration, providing an interactive and immersive learning experience.

ABSTRACT

Remote collaboration in mixed reality (MR) facilitates interaction across distributed locations; however, traditional approaches often fail to provide effective guidance for novices operating physical tools. While MR-based collaboration improves spatial awareness and communication by integrating real-world objects into virtual environments, existing systems struggle to fully replicate functional affordances, thereby limiting the effectiveness of remote instruction. This paper (i) defines novel user scenarios involving the functional operation of physical objects and delineates the corresponding user and system requirements, (ii) introduces **CRAFT** (Context-Responsive Affordance Framework for Teleinteraction), a framework that dynamically adapts virtual object affordances based on user interactions utilizing Vision-Language Models (VLMs), and identifies key technical challenges, and (iii) examines the broader landscape of affordance generation in MR, emphasizing the need for future research on dynamic, interaction-aware object virtualization to enhance remote collaboration.

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CCS CONCEPTS

• **Human-centered computing → Mixed / augmented reality**.

KEYWORDS

Mixed Reality, Teleinteraction, Context-aware System, Multimodal Interaction, Interactive System

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1 INTRODUCTION

Remote collaboration in mixed reality allows users to interact across distinct locations and is widely applied in education, craftsmanship, and equipment training. However, when novices learn to operate physical tools, text-based instructions or digitized materials detached from the physical context provide insufficient guidance. Even virtual overlays in remote assistance fail to provide precise, hands-on instruction for physical operations.

MR-based collaboration that integrates real-world objects into a shared virtual environment offers a promising solution. By improving spatial awareness and facilitating intuitive communication of procedural information, such systems enhance the effectiveness of remote guidance. Consider a scenario where a novice YouTuber (depicted in orange in Figure 1) struggles to extend the legs of a camera tripod in their physical space. An expert YouTuber (depicted in blue) can join the novice's environment virtually, providing real-time, immersive assistance. Even in the absence of a similar tripod in their own space, the expert can virtualize the novice's tripod as a digital object, manipulating it to demonstrate the correct operation.

However, a critical challenge arises: the functional properties of objects may not be fully preserved in virtualization, reducing the effectiveness of remote guidance. For instance, if the virtual tripod lacks extendability or has a restricted maximum height, it fails to accurately reflect the real-world tripod, making it harder for the novice to follow instructions intuitively. This limitation stems from the fact that current virtualization techniques struggle to fully replicate the affordances of physical objects. Affordance refers to the interactive properties that enable users to intuitively perceive and engage with objects. While virtual representations retain perceptual affordances based on shape and visual structure, they often lack the functional properties essential for operating complex tools, such as camera tripods or cooking utensils. Advanced functionalities, such as extending a tripod's legs in multiple stages, are difficult to predefine due to the resource-intensive design process, resulting in limited interactivity in virtual environments. This gap presents a significant challenge in remote spatial collaboration, particularly when users require precise, hands-on guidance to interact effectively with physical objects.

We introduce **CRAFT** (Context-Responsive Affordance Framework for Teleinteraction), an innovative framework for on-demand affordance generation in virtualized objects. **CRAFT** adapts to user-involved interaction contexts and enables users to access timely and relevant interaction capabilities for virtual objects, thereby enhancing remote hands-on collaboration by responding to user needs.

Our approach uses Vision-Language Models (VLMs) to dynamically adapt affordances by interpreting both visual (e.g., user pose and gestures) and verbal inputs (e.g., user intent). By integrating these modalities, our method generates object scripts on demand, modifying virtual object behaviors and structures without predefined interactions. This enables users to intuitively access needed functionalities, enhancing flexible and adaptive mixed reality interactions.

Our contributions are as follows:

- We introduce novel user scenarios for MR-based remote collaboration that involve the functional operation of physical objects, and outline the corresponding user-side and system-side requirements.
- We present **CRAFT**, a dynamic framework for affordance generation, and assess the maturity of the key technologies necessary to realize these scenarios in alignment with the identified requirements.
- We refine the task space for affordance generation, addressing the key technical challenges and complexities involved.

This paper does not aim to provide a complete solution but rather to frame the problem and set the stage for future research in object affordance modeling and generation.

2 RELATED WORKS

2.1 Remote Collaboration Systems in MR

Remote collaboration systems have been widely recognized for their effectiveness, leading to the development of platforms that enable intuitive and spatially aware interactions. Among these, spatial sharing applications aim to virtualize and share users' physical environments to (i) reduce cognitive load by allowing seamless navigation of physical and virtual spaces and (ii) enhance social presence through spatial alignment, improving immersion and coordination. For example, BeHere [18] enables remote experts in virtual reality to guide local workers in real time using 3D virtual replicas, avatars, and gesture sharing, enhancing task execution.

While such systems improve social presence and task efficiency, they remain limited in supporting interactive, hands-on collaboration, as they primarily facilitate one-way interactions with physical objects. This limitation restricts their effectiveness in scenarios requiring dynamic object manipulation bidirectionally.

To bridge this gap, we define a novel scenario and introduce a dynamic affordance generation framework that allows virtual objects to adapt their functionalities in real-time based on user interactions. By enabling richer and more interactive remote demonstrations, this approach significantly enhances the usability of remote collaboration systems.

2.2 Runtime Content Generation in MR

Runtime 3D content generation is a well-established research area within the extended reality (XR) field. Existing 3D content generation systems leverage user inputs and environmental data to dynamically generate 3D scenes, objects, and avatars within virtual environments [1, 5, 6, 10, 16].

Recent studies have explored the semantic-driven generation of virtual objects. Some studies leverage surrounding physical and virtual elements to inform the creation of 3D content. For instance, XR-Objects [6] identifies physical objects within a user's field of view and constructs an object-centered user interface based on their semantic properties. CAST [21] takes 2D image as an input, detects objects and analyze the inter-object relationship by leveraging LLM, to reconstruct the image as a 3D scene. On the other hand, some studies leverage human requests as input, capturing semantics to generate 3D content. LLMR [5] demonstrates the generation of virtual objects by analyzing both environmental semantics and user requests. Similarly, Syncity [7] showcases the creation of complex, coherent 3D worlds from user prompts.

Existing methods for modifying virtual objects mainly focus on surface-level edits, such as color changes, animation adjustments [5], and predefined interface elements [6], limiting dynamic interaction capabilities in complex scenarios. While LLMR [5] demonstrates the potential of large language models (LLMs) for enhancing 3D object interactivity, its application to object-level editing for advancing object-human interaction remains underexplored.

To address this gap, **CRAFT** explores the on-demand generation of functional affordances for virtual objects based on user interaction contexts, supporting more immersive and context-aware virtual experiences. This paper also evaluates the technical maturity of **CRAFT** as a foundational building block for future developments.

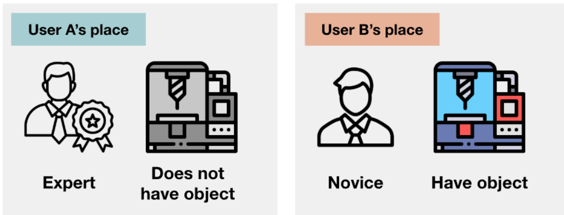

Figure 2: Scope specification of remote collaboration scenario targeting in CRAFT. User A (expert) is remotely located without access to the physical object, while User B (novice) has the object but lacks expertise. Through a virtual-augmented reality system, User A provides instructions, demonstrations, and corrective feedback to guide User B in operating the object efficiently.

2.3 Object Affordance Generation in MR

Affordance theory plays a central role in shaping contemporary XR interaction paradigms [2], with affordances categorized into perceptual, physical, and functional types, influencing how users interact with virtual objects.

Recent computational approaches to affordance generation include geometry-based mappings [2], vision-language correlations [4], and interaction-driven learning [11, 20]. While effective, these methods primarily focus on static affordances (e.g., grabbable or sittable).

To address limitations in static affordances, some methods have extended affordance generation to articulated objects, enabling interaction with object segments. In robotics, iterative exploration of articulated objects refines affordances through feedback [19], and XR applications use interactive perception to model object articulation from real-world images [9]. However, these methods are sensitive to uncertainty and error propagation and often struggle to integrate dynamic human expertise, limiting their effectiveness in real-world scenarios.

To overcome these challenges, we propose a dynamic, interaction-driven framework for affordance generation based on user motion demonstrations. Unlike previous methods, which rely on static semantics or physical properties, this framework incorporates real-time expert user demonstrations and Vision-Language Models (VLMs) to refine affordances dynamically, enhancing adaptability and enabling richer, more interactive XR collaboration.

3 USER SCENARIO

3.1 Specification of Scenario

- **Remote Collaboration in the Absence of a Physical Object**: This scenario involves two users: User A, an expert in the operation of a complex physical object, but without direct access to the object, and User B, a novice with physical access to the object.
- **Mixed Reality Interaction Dynamics**: User A participates in the interaction within a fully virtual environment, perceiving User B's space and the object virtually. In contrast, User B engages in a mixed-reality environment that integrates both physical and virtual elements, including the avatar of User A and supplemental virtual objects for guidance.
- **Physical Object as the Central Context**: The interaction centers around a physical object with functional attributes extending beyond mere manipulation or grasping. This complexity necessitates precise, contextualized instructional guidance to ensure effective operation.
- **Complex Expert-Novice Instructional Setting**: User A, as the expert, provides instruction to User B on how to operate the object. Traditional methods, such as verbal instructions or pointing gestures, may be fragmented and inefficient, especially when the operation entails multiple steps or when misunderstandings arise. In time-sensitive tasks, multi-turn instructions may be impractical due to the urgency of immediate action.

3.2 Example Scenarios

Scenario 1: Industrial Machinery Troubleshooting
A field technician (User B) in a remote plant faces a malfunctioning robotic assembly and lacks the expertise to repair it. A senior engineer (User A), located remotely, assists through a virtual-augmented reality interface. The technician sees the engineer's avatar and virtual annotations guiding them to inspect or adjust components. Complex troubleshooting, such as recalibrating sensors or adjusting mechanical parts, requires precise visual and gestural guidance.

Scenario 2: Remote Training in Culinary Arts
A professional chef (User A) remotely teaches a culinary student (User B) advanced knife techniques and plating. The student follows the chef's real-time demonstrations of cutting angles and hand positioning. While verbal instructions like "slice at a 45-degree angle" provide some clarity, the lack of tactile feedback can cause inefficiencies. By incorporating dynamic visual markers and hand-tracking, the system ensures more precise replication of the chef's movements, reducing ambiguity and enhancing the learning experience.

3.3 Requirements

User-side requirements that are derived from user scenario in Section 3.1, and corresponding system-side requirements are described in Figure 3.

4 CRAFT

In this section, we elaborate on the role of each components within the overall system architecture and present the field's technical maturity with foundational researches.

4.1 Mixed-Reality Engine

The Mixed-Reality Engine enables real-time spatial mapping, object tracking, and virtual overlays, blending physical and digital environments. Frameworks like Unity XR [17] and Unreal Engine [8] provide core functionalities for spatial awareness, gesture recognition, and interactive object placement in mixed reality.

4.2 Object Replicator

This module virtualizes physical objects with high fidelity, capturing their geometry, texture, and perceptual affordances to enable seamless XR integration. Advances in Neural Radiance Fields

User-side Requirement	System-side Requirement	System Component	Technical Maturity
UR1. User can engage in real-time multimodal interaction with registered objects	**SR1**. Mixed-Reality Integration	Mixed-Reality Engine, Interaction Cue Analyzer	**Mature**
UR2. User can register new physical objects as central interaction elements	**SR2**. Physical object virtualization	Object Replicator	**Developing**
UR3. User can express intent through speech, gestures, and object interactions to the system	**SR3**. Multimodal Input Recognition	Object-of-Interest Selector, Interaction Cue Analyzer	**Developing**
UR4. User can analyze possible affordances of an existing object	**SR4**. System identifies and evaluates potential affordances for the object	Affordance Analyzer	**Developing**
UR5. User can dynamically generate the most contextually relevant affordance for an object	**SR5**. System generates the best affordance that fits the current interaction context	Affordance Generator	**Emerging**

Figure 3: Mapping between user-side requirements, system-side requirements, and actual system components. The Implementation column indicates whether existing work supports the corresponding system components. While some systems enable the Affordance Generator, they exhibit limitations, as discussed in Section 2.3.

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(NeRFs) [12] and photogrammetry-based reconstruction [14] have significantly improved object digitization, allowing for highly detailed virtual replicas. However, replicating physical objects into a 3D interactable object in XR has remaining challenges.

4.3 Object-of-Interest Selector

This module identifies the user's focus by capturing the position and pose of both virtual and physical hands. Previous research, such as GradualReality [15], uses hand proximity and dynamic blending to select target objects. A comparative analysis of existing techniques will inform the system's design, improving accuracy, responsiveness, and adaptability in object selection.

4.4 Interaction Cue Analyzer

This module interprets user input across verbal commands, hand gestures, and object manipulations, to infer user intent. By analyzing pointing gestures, motion demonstrations, and speech-based interactions, this module refines affordance selection and ensures contextual alignment. Prior works leveraging user verbal cues, or user body part position and gesture [13] tailored to the system goal provides a good building block for robust intent prediction. However, user motion as an interaction cue for functional affordance demonstration remains underexplored. **CRAFT** seeks to bridge this gap by utilizing motion-based affordance inference.

4.5 Affordance Analyzer

This module identifies available affordances by integrating user intent and object positional context. Building on insights from grounded affordance learning [20, 22], this module dynamically evaluates affordances to ensure intuitive and responsive interactions. It prioritizes affordances based on hand proximity, gaze direction, and inferred task objectives, aligning system responses with user expectations.

4.6 Affordance Generator

This module dynamically modifies virtual objects based on analyzed affordances, enabling contextual interactivity ranging from basic grasping to complex manipulations. To incorporate affordances, objects may be articulated—segmented while preserving their overall shape—based on semantic roles, as explored in Piqa [3]. **CRAFT** aims to leverage this approach, enabling seamless affordance adaptation while maintaining object integrity.

5 DISCUSSION POINTS

The task of object affordance generation can be conceptualized through several distinct subtasks:

Enhancing Interactivity of Rigid Objects. Enabling basic manipulations like grabbing and dragging through colliders and user action triggers (e.g., gestures, clicks), allowing direct interaction within the virtual environment.

Decomposing Objects into Multipart Interactions. Dividing objects into parts or overlays to maintain interactivity without altering their overall structure, such as making a valve rotatable by separating it from the main component.

Defining Joint Constraints for Multipart Objects. Establishing physical or logical constraints between parts to control interactions, ensuring functionality, such as making a valve rotatable but not pullable or a tripod extendable but not foldable.

Discussion Point 1. How can we match user inputs to leverage AI capacities for each subtask? For complex tasks like joint constraints, do we need more detailed inputs like user gestures? How should these inputs be utilized?

Discussion Point 2. In LLM-based generation, modifying existing structures while preserving essential details is often more challenging than generating new ones. Similarly, in affordance generation, articulating objects and integrating new affordances without disrupting their original form or function presents a significant challenge. Can solutions from the LLM field offer insights or methods to address this issue?

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